

## // SANITY CHECK

```
docker version      # client + server
docker run --rm hello-world
```

## // IMAGES = LAYERS

```
docker pull postgres:18
docker history postgres:18  # see the layers
docker images
docker rmi <image>
```

## // COMPOSE · THE COURSE STACK

All infra lives in `infra/compose.yaml`. Run from repo root.

```
# start postgres + kafka + kafka-ui
docker compose -f infra/compose.yaml \
  up -d postgres kafka kafka-ui

docker compose -f infra/compose.yaml ps
# all Up, postgres healthy
```

## // EXEC INTO A CONTAINER

```
docker compose -f infra/compose.yaml \
  exec postgres psql -U orders -d orders
```

## // TEAR DOWN

```
docker compose ... down      # keep volumes
docker compose ... down -v   # wipe volumes
```

Named volume `pgdata` outlives `down` ; only `down -v` deletes your data.

## // NETWORKING

```
in-network  postgres:5432 / kafka:9092
from laptop localhost:5432 / :29092
```

A service reaches another by **service name**, not localhost.

## // PORT OVERRIDE

```
export POSTGRES_HOST_PORT=5433
docker compose ... up -d postgres
```

## // BUILD &amp; RUN AN IMAGE

```
docker build -f src/main/docker/Dockerfile.jvm \
  -t quarkus/kyc-service-jvm .
docker run -i --rm -p 8080:8080 \
  quarkus/kyc-service-jvm
```

## // MULTI-STAGE DOCKERFILE

`apps/kyc-service/Dockerfile` — read this one.

```
# syntax=docker/dockerfile:1
FROM maven:3.9-eclipse-temurin-25 AS build
COPY pom.xml .
RUN --mount=type=cache,target=/root/.m2 \
  mvn -B -ntp -DskipTests \
  dependency:go-offline
COPY src ./src
RUN ... mvn -B clean package

FROM eclipse-temurin:25-jre AS runtime
RUN useradd app
COPY --from=build .../lib/ ./lib/
COPY --from=build .../*.jar ./
COPY --from=build .../app/ ./app/
COPY --from=build .../quarkus/ ./quarkus/
USER app
EXPOSE 8080
ENV QUARKUS_PROFILE=prod
ENTRYPOINT ["java","-jar","/work/quarkus-run.jar"]
```

Copy `pom.xml` before `src` so the dependency layer caches across rebuilds.

## // EVERYDAY

```
docker ps          # running
docker logs -f <c> # follow
docker stats       # live usage
docker system prune # reclaim space
```